

Quantum Error Correction

Part 1: From Tanner Codes to Good Quantum LDPC Codes

Quantum Existential Crisis (QEC)

Summer 2026

Classical Coding Theory

A binary linear code is a subspace $C \subseteq \mathbb{F}_2^n$. It has parameters $[n, k, d]$, where

$$n = \text{codeword length}, \quad k = \text{message length}, \quad d = \min_{0 \neq c \in C} |c|.$$

A linear code can be specified by a parity-check matrix H :

$$C = \ker H = \{x \in \mathbb{F}_2^n \mid Hx = 0\}, \quad \dim C = k.$$

Example

Repetition code: $0 \mapsto 0000$, $1 \mapsto 1111$

Parity-check code: $0101 \mapsto 0101\mathbf{0}$, $1110 \mapsto 1110\mathbf{1}$

A family is *asymptotically good* if

$$k = \Theta(n), \quad d = \Theta(n).$$

It is Low Density Parity Check(LDPC) if H has bounded row and column weights.

Classical Tanner codes

A Tanner code is built from:

- a D -regular bipartite graph $\Gamma = (L \sqcup R, E)$, $|L| = |R| = m$, $|E| = Dm$,
- a fixed $[D, k_0, d_0]$ local code $C_0 \subseteq \mathbb{F}_2^D$,
- for any $F \subset E$, $\mathbf{1}_F \in C$ requires $x|_{E(v)} \in C_0$.

Why is Tanner code good?

Theorem (Expander Mixing Lemma)

For any $S \subseteq L, T \subseteq R$,

$$\left| e(S, T) - \frac{D}{m} |S| |T| \right| \leq \lambda_2 \sqrt{|S| |T|}.$$

$$d_0 \sqrt{|S| |T|} \leq |F| \leq \frac{D}{m} |S| |T| + \lambda_2 \sqrt{|S| |T|} \implies |F| \geq \frac{d_0(d_0 - \lambda_2)}{D} m.$$

Takeaway from Tanner Code:

linear distance local code + global expander graph $\implies$ linear distance.

Quantum errors are still binary

A quantum code is a subspace

$$Q \subseteq (\mathbb{C}^2)^{\otimes n}.$$

For one qubit, the Pauli errors are $\{I, X, Z, Y = iXZ\}$. Ignoring global phases, every n -qubit Pauli errors can be written as

$$X^{\vec{a}} Z^{\vec{b}}, \quad \vec{a}, \vec{b} \in \mathbb{F}_2^n.$$

Example

A *CSS stabilizer code* is specified by two binary matrices H_X, H_Z , whose code space is

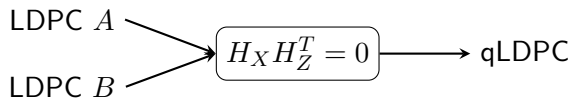
$$Q = \left\{ |\psi\rangle : X^{\vec{a}} |\psi\rangle = |\psi\rangle \ \forall \vec{a} \in \text{row}(H_X), \ Z^{\vec{b}} |\psi\rangle = |\psi\rangle \ \forall \vec{b} \in \text{row}(H_Z) \right\}.$$

All stabilizers need commute:

$$X^{\vec{a}} Z^{\vec{b}} = Z^{\vec{b}} X^{\vec{a}} \iff \vec{a} \cdot \vec{b} = 0 \iff H_X H_Z^T = 0.$$

Finding qLDPC

Basic Framework (Calderbank, Shor, Steane, 1996):



Hypergraph product (Tillich, Zémor, 2009)

Take two classical parity-check matrices A, B . Define

$$H_X = (A \otimes I; ; I \otimes B^T), \quad H_Z = (I \otimes B; ; A^T \otimes I).$$

Then automatically $H_X H_Z^T = 0$. If A, B are LDPC, so is the resulting quantum code.

If the two classical codes have size L , then

$$n = \Theta(L^2), \quad d = \Theta(L) = \Theta(\sqrt{n}).$$

First good qLDPC

Theorem (Panteleev–Kalachev, 2021)

There exists an explicit family of qLDPC codes with $k = \Theta(n)$, $d = \Theta(n)$.

Main idea.

- Start with two Tanner codes have a free G -action on their coordinates. This is obtained by a particular Cayley graph with group G . Their parity-check matrices can be written as constant-size matrices over $R = \mathbb{F}_2[G]$
- The *lifted product* is the tensor product $A \otimes_R B$. Since $R \otimes_R R \cong R$, only one copy of the G -coordinate remains, and hence $n = \Theta(|G|)$.
- The two local codes are chosen to be *product-expanding*: a local $w \times w$ pattern cannot become sparse through cancellation between its codeword-rows and codeword-columns.
- Result:

local product expansion + global Cayley expansion $\implies d = \Theta(n)$.

Development of qLDPC Codes

Work	Contribution
Panteleev–Kalachev (2021)	First good qLDPC: $k, d = \Theta(n)$.
Leverrier–Zémor (2022)	Cleaner Tanner-style construction.
Dinur–Hsieh–Lin–Vidick (2022)	Good qLDPC with linear-time decoding.
Hsieh–Lubotzky–Mohanty–Reiner–Zhang (2025)	Explicit constant-degree lossless expanders.
Wills–Lin–Zhang–Hsieh (2026)	Good quantum codes with fast encoding/decoding.

Remains open:

Are there quantum code with:

- Constant rate, linear distance
- Fast encoding/decoding algorithm
- Explicit construction
- Fault tolerance

References

- Panteleev and Kalachev, *Asymptotically Good Quantum and Locally Testable Classical LDPC Codes*, arXiv:2111.03654.
- Tillich and Zemor, *Quantum LDPC codes with positive rate and minimum distance proportional to the square root of the blocklength*, arXiv:0903.0566.
- Leverrier and Zemor, *Quantum Tanner codes*, arXiv:2202.13641.
- Leverrier and Zemor, *Decoding quantum Tanner codes*, arXiv:2208.05537.
- Dinur, Hsieh, Lin, and Vidick, *Good Quantum LDPC Codes with Linear Time Decoders*, arXiv:2206.07750.
- Lin and Hsieh, *Good quantum LDPC codes with linear time decoder from lossless expanders*, arXiv:2203.03581.
- Hsieh, Lubotzky, Mohanty, Reiner, and Zhang, *Explicit Lossless Vertex Expanders*, arXiv:2504.15087.
- Wills, Lin, Zhang, and Hsieh, *Linear-Time Encodable and Decodable Quantum Error-Correcting Codes*, arXiv:2603.04543.

Part II: A "Geometric" qLDPC Code

What if, instead of optimizing asymptotic parameters, we insist in geometric locality?